Mikolaj Kubowicz
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Q3.1a
R1 has two paths to network 10.3.3.0/24 in the routing table and two successors in the EIGRP topology table. Both routes have the same EIGRP metric of 131072, so EIGRP is using both paths. In [90/131072], 90 is the administrative distance for an internal EIGRP route, and 131072 is the EIGRP metric used to determine the best path. THe EIGRP topology table also shows a feasible distance of 131072

Q3.1b
After changing the WAN bandwidths to 100 Mbps, R1 still has two paths to 10.3.3.0/24 in both the routing table and the EIGRP topology table. The EIGRP metric increased from 131072 to 154112 because bandwidth is part of EIGRP's metric calculation. Both paths still have equal metrics, so both remain successors.

Q3.2
After shutting down R2 G0/0, R1 lost the EIGRP path through R2. The route to 10.3.3.0/24 now has only one successor, through R4 at 10.1.1.4. This is different from before because R1 previously had two equal-cost paths. EIGRP automatically updated the routing table after the R2 path became unavailable.

Q3.3
After R2 G0/0 was brought back up, R1 formed a new EIGRP adjacency with R2 at 10.1.1.2. The debug messages showed EIGRP advertising routes and recalculating the routing table. The route to 10.3.3.0/24 through R2 had a metric of 134912, while the route through R4 had a metric of 154112. The R2 path had the lower metric because the R2-R3 link was changed to 400 Mbps while the R4-R3 link remained at 100 Mbps, so EIGRP preferred the path through R2. The route installed messages show EIGRP updating the routing table after the neighbor came back online.

Q3.4

R1 now has one path to the 10.3.3.0/24 network in the routing table and one successor in the EIGRP topology table. The path goes through R2 at 10.1.1.2 with a metric of 134912.

This is different from Q3.1, where R1 had two equal-cost paths to 10.3.3.0/24 through both R2 and R4. After changing the R2-R3 link bandwidth to 400 Mbps while the R4-R3 link remained at 100 Mbps, the path through R2 received the lower EIGRP metric and became the preferred path.

This differs from RIP because RIP chooses routes mainly by hop count, so changing interface bandwidth does not change the RIP metric. EIGRP uses bandwidth as part of its metric calculation, so changing bandwidth can change which path is preferred.

Q.3.5

After configuring variance 8 on R3, I saw several “route installed” messages. EIGRP installed additional unequal-cost paths in R3’s routing table. For example, R3 now had two paths to the 172.16.40.0/24 and 172.16.41.0/24 networks, with different metrics. R1 did not change and still had one preferred route to 10.3.3.0/24 through R2. This happened because variance 8 was configured only on R3, allowing R3 to use qualifying unequal-cost EIGRP paths.

Q.3.6

R3 currently has four routes in the 172.x.x.x range: 172.16.40.0/24, 172.16.41.0/24, 172.17.0.0/16, and 172.18.0.0/16. These routes are being learned through EIGRP from R1. The 172.17.0.0/16 and 172.18.0.0/16 routes came from the new Loopback17 and Loopback18 interfaces created on R1.

Q.3.7

I configured EIGRP summarization on R1 G0/0 using ip summary-address eigrp 888 172.16.0.0 255.252.0.0. After the change, R3 no longer showed the individual 172.16.x.x, 172.17.x.x, and 172.18.x.x routes. Instead, it showed one summarized EIGRP route, 172.16.0.0/14. This confirms that the route summarization worked.

Q.3.8

Yes, R1 now has a summary route for 172.16.0.0/14 pointing to Null0. This route was automatically created when the EIGRP summary address was configured. Null0 acts as a discard route. If R1 receives traffic for an address inside the summarized 172.16.0.0/14 range but does not have a more specific route for that destination, the traffic is dropped instead of being forwarded incorrectly. This helps prevent routing loops.

HOURS WORKED 4